

Session2: A Test of a Nonlinear Combination of Coefficients

Table 2 (Columns 1–4) in *Income and Democracy* (AER 2008)

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Session

1 Nonlinear inference for the implied cumulative (long-run) effect

This note summarizes the estimation logic behind Table 2, Columns (1)–(4) in Acemoglu, Johnson, Robinson, and Yared (2008), *Income and Democracy*. The empirical design is a dynamic panel with country fixed effects and year effects, estimated on five-year interval data (and variants), with standard errors clustered by country. A recurring object of interest is the *implied cumulative (long-run) effect of income* in a dynamic specification with persistence.

1.1 OLS baseline: conditional expectation and sampling error

Consider the linear model

$$y = X\beta + \varepsilon, \quad (1)$$

where we condition on X . The OLS estimator is

$$\begin{aligned} \hat{\beta} &= (X'X)^{-1}X'y \\ &= (X'X)^{-1}X'(X\beta + \varepsilon) \\ &= \beta + (X'X)^{-1}X'\varepsilon. \end{aligned} \quad (2)$$

Taking conditional expectations,

$$\mathbb{E}[\hat{\beta} \mid X] = \beta + (X'X)^{-1}X'\mathbb{E}[\varepsilon \mid X]. \quad (3)$$

Under the exogeneity condition $\mathbb{E}[\varepsilon \mid X] = 0$, we obtain

$$\mathbb{E}[\hat{\beta} \mid X] = \beta. \quad (4)$$

Moreover, the deviation from its conditional mean (sampling error) is

$$\begin{aligned} \hat{\beta} - \mathbb{E}[\hat{\beta} \mid X] &= \hat{\beta} - \beta \\ &= (X'X)^{-1}X'\varepsilon. \end{aligned} \quad (5)$$

1.2 Dynamic panel estimand: the implied cumulative effect

In `acemoglu2008income`, the baseline dynamic panel includes lagged democracy and lagged income. Abstracting from other controls for exposition, write

$$d_{it} = a d_{i,t-1} + g y_{i,t-1} + m_t + d_i + u_{it}, \quad (6)$$

where d_{it} is democracy, y_{it} is log income per capita, m_t are year effects, and d_i are country fixed effects.

Because democracy is persistent, the object reported in Table 2 is not only the short-run coefficient g , but also the implied cumulative (long-run) effect of a permanent change in income:

$$\psi \equiv \frac{g}{1-a}, \quad (\text{stability requires } |a| < 1). \quad (7)$$

This quantity is a *nonlinear function* of the estimated coefficients $(\hat{g}, \hat{a})$.

1.3 Why nlcom: nonlinear combination, delta method, and a Wald test

Let $\theta = (g, a)'$ and define the nonlinear map $f(\theta) = g/(1-a)$. Given an estimator $\hat{\theta}$ (from OLS/FE/IV/GMM), the plug-in estimate of the long-run effect is

$$\hat{\psi} = f(\hat{\theta}) = \frac{\hat{g}}{1-\hat{a}}. \quad (8)$$

Stata's `nlcom` computes $\hat{\psi}$ and its standard error using a first-order delta method:

$$\widehat{\text{Var}}(\hat{\psi}) \approx \nabla f(\hat{\theta})' \widehat{\text{Var}}(\hat{\theta}) \nabla f(\hat{\theta}), \quad (9)$$

where $\widehat{\text{Var}}(\hat{\theta})$ is the (cluster-)robust covariance matrix from the underlying estimator. For $f(g, a) = g/(1-a)$,

$$\nabla f(\theta) = \begin{pmatrix} \frac{\partial f}{\partial g} \\ \frac{\partial f}{\partial a} \end{pmatrix} = \begin{pmatrix} \frac{1}{1-a} \\ \frac{g}{(1-a)^2} \end{pmatrix}. \quad (10)$$

The reported p -value for `cum_eff` corresponds to a Wald test of the nonlinear restriction

$$H_0 : \psi = 0 \quad \text{vs.} \quad H_1 : \psi \neq 0, \quad (11)$$

using the test statistic $z = \hat{\psi}/\widehat{\text{se}}(\hat{\psi})$ (asymptotically standard normal under H_0).

Stata snippet (for Table 2 long-run effect)

```
* Long-run (cumulative) effect: psi = g/(1-a)
nlcom _b[L.lrgdpch]/(1-_b[L.fhpolrigaug])
```

2 Context, Notation, and Target Estimand

Panel structure. Let i index countries and t index time (in five-year steps for the baseline panel). Define:

- d_{it} : democracy measure (e.g., Freedom House Political Rights index, augmented as in Barro).
- y_{it} : log GDP per capita.
- m_t : time (year) effects capturing common shocks.
- d_i : country fixed effects capturing time-invariant country characteristics.
- u_{it} : residual term.

Dynamic panel specification (paper's baseline equation). The main regression (paper's Eq. (1) in Section II) can be written as:

$$d_{it} = a d_{i,t-1} + g y_{i,t-1} + x'_{i,t-1} b + m_t + d_i + u_{it}, \quad (12)$$

where a captures persistence in democracy, and g is the short-run effect of income.

Implied cumulative (long-run) effect of income. Because democracy is persistent, a permanent increase in income affects democracy over multiple periods. Under stability ($|a| < 1$), the long-run (cumulative) effect of income is:

$$\text{LR effect} \equiv \frac{g}{1-a}. \quad (13)$$

In the paper’s Table 2 notes, the “Implied cumulative effect of income” is computed as the coefficient on log GDP per capita at $t-1$ divided by $(1 - \text{democracy}_{t-1})$, and the bracketed value is the p -value from a nonlinear test of this quantity.

3 A Minimal OLS Derivation: Conditional Expectation and Sampling Error

Consider the linear model

$$y = X\beta + \varepsilon, \quad (14)$$

where X is treated as fixed in repeated samples (or we condition on X).

OLS estimator and decomposition.

$$\hat{\beta} = (X'X)^{-1}X'y \quad (15)$$

$$= (X'X)^{-1}X'(X\beta + \varepsilon) \quad (16)$$

$$= \beta + (X'X)^{-1}X'\varepsilon. \quad (17)$$

Conditional expectation (unbiasedness under exogeneity). Taking conditional expectation given X :

$$\mathbb{E}[\hat{\beta} \mid X] = \mathbb{E}[\beta + (X'X)^{-1}X'\varepsilon \mid X] \quad (18)$$

$$= \beta + (X'X)^{-1}X'\mathbb{E}[\varepsilon \mid X]. \quad (19)$$

If the exogeneity condition holds:

$$\mathbb{E}[\varepsilon \mid X] = 0, \quad (20)$$

then:

$$\mathbb{E}[\hat{\beta} \mid X] = \beta. \quad (21)$$

Deviation from its conditional mean. Using (17) and $\mathbb{E}[\hat{\beta} \mid X] = \beta$:

$$\hat{\beta} - \mathbb{E}[\hat{\beta} \mid X] = \hat{\beta} - \beta \quad (22)$$

$$= (X'X)^{-1}X'\varepsilon. \quad (23)$$

4 Table 2, Columns (1)–(4): Principles and Estimation Steps

The paper’s Table 2 explicitly labels Columns (1)–(4) as: Pooled OLS, Fixed Effects OLS, Anderson–Hsiao IV, and Arellano–Bond GMM (for the baseline five-year panel).

4.1 Column (1): Pooled OLS (Baseline in the Literature)

Idea. Estimate the dynamic specification using pooled OLS with year dummies, omitting country fixed effects d_i . This corresponds to Eq. (12) without d_i :

$$d_{it} = a d_{i,t-1} + g y_{i,t-1} + x'_{i,t-1} b + m_t + u_{it}.$$

Pooled OLS is a natural baseline but can be biased if d_i is correlated with $y_{i,t-1}$ or other regressors.

Implementation features.

- Include year dummies (m_t) to absorb common shocks.
- Cluster standard errors by country to allow within-country serial correlation and heteroskedasticity.
- Compute the implied cumulative effect $g/(1-a)$ via a nonlinear combination of estimates.

4.2 Column (2): Fixed Effects OLS (Country Dummies / Absorbed FE)

Idea. Control for time-invariant country heterogeneity by including country fixed effects d_i . This removes bias driven by omitted time-invariant factors correlated with income or lagged democracy.

Dynamic FE caveat (Nickell bias). With a lagged dependent variable, the within (FE) transformation makes the transformed regressor $(d_{i,t-1} - \bar{d}_i)$ mechanically correlated with the transformed error $(u_{it} - \bar{u}_i)$ when T is not large, generating the classic finite- T dynamic panel bias (Nickell bias). This motivates moving to IV/GMM approaches.

4.3 Column (3): Anderson–Hsiao (1982) Difference-IV (2SLS on the Differenced Equation)

Step 1: Time-difference Eq. (12) to remove d_i . Differencing yields the paper’s Eq. (2):

$$\Delta d_{it} = a \Delta d_{i,t-1} + g \Delta y_{i,t-1} + \Delta x'_{i,t-1} b + \Delta m_t + \Delta u_{it}. \quad (24)$$

Step 2: Address endogeneity created by $\Delta d_{i,t-1}$. Even after differencing, OLS on (24) is inconsistent because $\Delta d_{i,t-1}$ contains $u_{i,t-1}$ while $\Delta u_{it} = u_{it} - u_{i,t-1}$ also contains $u_{i,t-1}$.

Step 3: Use lagged levels as instruments (double lag). Under the key condition that the original residual u_{it} has no serial correlation (equivalently, no second-order serial correlation in Δu_{it}), the paper instruments:

$$\Delta d_{i,t-1} \text{ with } d_{i,t-2}, \quad \Delta y_{i,t-1} \text{ with } y_{i,t-2}.$$

This is a consistent IV/2SLS estimator but uses only a limited subset of available moment conditions.

4.4 Column (4): Arellano–Bond (1991) Difference-GMM (More Efficient)

Motivation. Anderson–Hsiao uses a small number of instruments. Under the same no-further-serial-correlation condition, all deeper lags of democracy are also valid instruments for the differenced equation. Arellano–Bond (1991) builds a GMM estimator that exploits *all* such valid moment conditions, which is more efficient when valid.

Core moment conditions (difference-GMM). For $s \geq 2$ and relevant t , valid instruments satisfy:

$$\mathbb{E}[d_{i,t-s} \cdot \Delta u_{it}] = 0,$$

and analogously for other predetermined/endogenous regressors under their respective timing assumptions.

Diagnostics enabled by multiple instruments. Because difference-GMM is overidentified, it naturally supports:

- An AR(2) test on differenced residuals, probing whether the no-serial-correlation assumption needed for lag instruments is violated.
- A Hansen J test of overidentifying restrictions, probing the joint validity (exogeneity) of the instrument set.

Why not system-GMM here? The paper notes that additional level-equation moment conditions (system-GMM) would require time-differenced instruments to be orthogonal to country fixed effects, which is not appealing in this setting (e.g., five-year income growth is unlikely to be orthogonal to the democracy country fixed effect). Hence the focus remains on difference-GMM.

5 Stata Code Skeleton for Columns (1)–(4) and the Long-Run Effect

Packages (one-time)

```
* If needed:
ssc install ivreg2, replace
ssc install xtabond2, replace
ssc install estout, replace
```

Setup

```
use "five-years_Acemoglu_2008_AER.dta", clear
xtset code_numeric year_numeric
```

Helper: store implied cumulative (long-run) effect and its p-value

```
cap program drop add_cum
program define add_cum
    * After nlcom, r(table)[1,1] is the estimate and r(table)[4,1] is the p-value
    local cum_eff = r(table)[1,1]
    local pval    = r(table)[4,1]

    local cum_eff : di %4.3f 'cum_eff'
    local pval    : di %4.2f 'pval'
    local pval = "['pval']"

    estadd local cum_eff    "'cum_eff'"
    estadd local cum_eff_p "'pval'"
end
```

Column (1): Pooled OLS (POLS)

```
reg fhpolrigaug L.fhpolrigaug L.lrgdpch i.year if sample==1, cluster(code)

* Long-run effect:  $g/(1-a)$ 
nlcom _b[L.lrgdpch]/(1-_b[L.fhpolrigaug])
add_cum
est store m1
```

Column (2): Fixed Effects OLS (FE)

```
areg fhpolrigaug L.(fhpolrigaug lrgdpch) i.year if sample==1, absorb(code) cluster(code_num)

nlcom _b[L.lrgdpch]/(1-_b[L.fhpolrigaug])
add_cum
est store m2
```

Column (3): Anderson–Hsiao Difference-IV (AH)

```
ivreg2 D.fhpolrigaug ///
      (LD.(fhpolrigaug lrgdpch) = L2.(fhpolrigaug lrgdpch)) ///
      i.year if sample==1, cluster(code)

nlcom _b[LD.lrgdpch]/(1-_b[LD.fhpolrigaug])
add_cum
est store m3
```

Column (4): Arellano–Bond Difference-GMM (AB)

```
xtabond2 fhpolrigaug L.(fhpolrigaug lrgdpch) i.year if sample==1, ///
      gmm(L.(fhpolrigaug)) ///
      iv(i.year) iv(L2.lrgdpch, passthru) ///
      noleveleq robust

nlcom _b[L.lrgdpch]/(1-_b[L.fhpolrigaug])
add_cum
est store m4
```

Table output (example)

```
local mttitle "POLS FE AH AB"
local model "m1 m2 m3 m4"
local xx "L.fhpolrigaug L.lrgdpch LD.fhpolrigaug LD.lrgdpch"

esttab 'model', mttitle('mttitle') nogap compress ///
      b(%5.3f) se(%5.3f) brackets ///
      keep('xx') varwidth(16) ///
      scalar(hansenp ar2p cum_eff cum_eff_p N N_clust r2) ///
      sfmt(%4.2f %4.2f %4.2f %40.2f)
```

References

- [1] Acemoglu, Daron, Simon Johnson, James A. Robinson, and Pierre Yared. "Income and Democracy." *American Economic Review* 98(3) (2008): 808–842. DOI: 10.1257/aer.98.3.808. URL: <http://www.aeaweb.org/articles.php?doi=10.1257/aer.98.3.808>.